• Organize a scientific research process to solve a real world problem.	**1. Introduction to Research (5 hrs)** 1.1. Research: Objectives and Characteristics 1.2. Types of Research 1.3. The 6 P's of Research 1.4. Research Process 1.5. Research in Computer Science and Engineering

1. Explain the **6 P's of Research.** Illustrate how each stage contributes to the success of a research study with examples from IT research.

The **6 P's of Research** provide a comprehensive framework for understanding the essential components and stages of a successful research study. This model includes **Purpose, Paradigm, Process, Participants, Products, and Presentation**.

1. Purpose

The **Purpose** refers to the core reason for undertaking the study, the specific topic of interest, and the intended objectives or research questions.

- **Contribution to Success:** A clearly defined purpose ensures the research is worthwhile and provides a roadmap for the investigation; without it, research is unlikely to be high-quality.
- **IT Example:** In computer engineering, the purpose often involves **innovation** (developing novel algorithms) or **optimization** (improving existing systems to make them more secure). For instance, a study may aim to discover the most effective phishing website detection techniques to assist security managers.

2. Paradigm

A **Paradigm** is a shared way of thinking, a model, or a philosophical framework that guides how the researcher perceives reality and knowledge.

- **Contribution to Success:** It provides the underlying assumptions for the study. For example, the **Positivism** paradigm emphasizes objectivity and measurable facts, while **Interpretivism** focuses on understanding subjective human experiences.
- **IT Example:** The **survey strategy** in IT research is heavily associated with the **positivism** paradigm because it seeks to identify patterns and generalizations across large populations.

3. Process

The **Process** is the sequence of activities performed during the study, including establishing a conceptual framework, selecting a strategy (e.g., experiment or case study), and choosing data analysis methods.

- **Contribution to Success:** Following a systematic process ensures the research is valid, rigorous, and acknowledges its own boundaries and limitations.
- **IT Example:** A **Systematic Literature Review (SLR)** in IT follows a strict, rule-based process involving the design of research questions, searching specific electronic databases (like IEEE Xplore and ACM Digital Library), and applying strict inclusion/exclusion criteria.

4. Participants

Participants include everyone directly involved (such as interviewees or observed subjects) and those indirectly involved (such as editors, the academic community, and end-users).

- **Contribution to Success:** Success depends on dealing with all participants **legally and ethically**, ensuring rights like **informed consent** and **data privacy** are protected to maintain the study's integrity.
- **IT Example:** In a study on user satisfaction with an automation system, participants might be selected via **purposive sampling**. Researchers must ensure that any sensitive personal information collected is stored securely and handled with care.

5. Products

Products are the tangible and intangible outcomes of the research, representing the researcher's contribution to the knowledge frontier.

- **Contribution to Success:** Products must answer the original research questions and provide evidence-based conclusions that may lead to further investigation.
- **IT Example:** Products can include a final **thesis**, a peer-reviewed **journal article**, a **technical report** detailing research findings, or even a computer-based product like a new phishing detection tool.

6. Presentation

Presentation is the final stage, focusing on effectively communicating and disseminating the research to the academic community and stakeholders.

- **Contribution to Success:** Professional presentation is vital; if findings are not conveyed clearly through accepted academic styles, the audience may assume the entire research project was conducted unprofessionally.
- **IT Example:** IT research is often presented through **conference talks**, **thesis defenses**, or **software demonstrations**. Additionally, **infographics** are used to simplify

complex data—such as visualising phishing trends—making it accessible and engaging for different audiences.

2. “Research is the systematic approach to obtaining and confirming new and reliable knowledge.” Justify this statement and explain the objectives and characteristics of effective research.

The statement that **research is the systematic approach to obtaining and confirming new and reliable knowledge** is justified by the fact that research follows an orderly, sequential procedure rather than relying on accidental discovery. It is considered **"systematic"** because it is a planned and managed process involving a series of steps to ensure the information generated is credible. The goal of **"obtaining new knowledge"** is central to research, which seeks to create original insights that were not previously known and to extend the current "knowledge frontier". Furthermore, the process of **"confirming"** knowledge involves verifying, developing, and testing theories or hypotheses to ensure findings are accurate. For knowledge to be **"reliable,"** research must be grounded in empirical evidence, use precise measurements, and undergo rigorous analysis to minimize errors in interpretation.

Objectives of Research

The primary objectives of research are broad and vary depending on the field, but generally include:

- **Gaining Insight:** To achieve new familiarity with a phenomenon or reach an unexplained horizon of knowledge.
- **Accurate Portrayal:** To accurately describe the characteristics of a specific individual, group, or situation.
- **Determining Associations:** To identify the frequency with which something occurs or its association with other variables.
- **Discovering Truth:** To find facts and truth through a scientific structure of investigation.
- **Theory Development:** To put forward entirely new theories or resolve existing contradictions in a field of study.
- **Verification:** To test old conclusions with new data or find new conclusions using old data.
- **Innovation and Optimization:** In technical fields like computer engineering, objectives focus on developing novel algorithms (innovation) and improving existing processes to make them faster or more secure (optimization).
- **Problem Solving:** To address real-world challenges in areas such as healthcare, finance, and education through computational or scientific methods.

Characteristics of Effective Research

Effective research is defined by several key characteristics that ensure its quality and scientific rigor:

- **Empirical:** It is firmly based on direct observations or experimentation.
- **Systematic:** It follows an orderly and sequential procedure, often referred to as the research cycle.
- **Controlled:** All variables are kept constant except for those specifically being tested or experimented upon.
- **Hypothesis-Driven:** It employs testable statements to guide the investigative process.
- **Analytical:** It requires critical analysis of all data to ensure there are no errors in interpretation.
- **Objective and Unbiased:** All findings and conclusions must be logically based on empirical evidence, free from personal bias.
- **Quantitative/Statistical:** It often transforms data into numerical measures that can be treated statistically for more precise results.
- **Circular:** Research is an ongoing process that frequently leads to new questions, starting the cycle over again.
- **Creative:** While logical, the process also requires imagination, intuition, and initiative to discover new patterns and relationships.

3. How is research defined and conceptualized within the field of computer engineering? Explain the model of the research process.

Within the field of computer engineering, research is conceptualized as a **systematic investigation** designed to develop or contribute to **generalisable knowledge**. It is fundamentally viewed as the act of **extending the "knowledge frontier"** by building upon existing work to solve real-world problems. Specifically, research in this domain focuses on **innovation**, such as developing novel algorithms and architectures, and **optimisation**, which involves making existing technologies faster, more efficient, or more secure. Furthermore, computer engineering research seeks to advance **automation** and address complex challenges in diverse areas like healthcare and finance using computational methods.

Model of the Research Process

The research process is defined as a planned, managed, and **creative sequence of activities**. It is often described as a **circular cycle** because the interpretation of findings frequently leads back to new questions, starting the process over again.

A standard model of this process typically involves the following seven steps:

1. **Identification of a general problem or question:** The process is initiated by a specific issue or a "gap" in existing knowledge.

2. **Literature review:** Researchers examine previous studies to understand what has been done and to refine their own focus.
3. **Specification of questions or hypotheses:** Broad problems are narrowed into testable predictions or specific interrogative statements.
4. **Determination of design and methodology:** A master plan is developed, identifying the **research strategy** (such as an experiment, case study, or simulation) and the **conceptual framework**.
5. **Data collection:** Using specific methods like surveys, interviews, or observations, the researcher generates the raw material required for analysis.
6. **Data analysis and presentation:** Collected information is prepared and analysed, often using **statistical methods** for quantitative data or thematic coding for qualitative data.
7. **Interpretation of findings:** The researcher draws conclusions, evaluates whether the results support the hypotheses, and identifies implications for practice or further study.

This process is guided by **philosophical paradigms**, such as **positivism** (focusing on measurable facts) or **interpretivism** (focusing on human experience), which shape how the researcher perceives reality and knowledge. Ultimately, the model functions as a rigorous framework to ensure that the generated information is credible and reliable.

4. Describe briefly with examples the role of the “6 P’s of Research” in shaping a research project.

The **6 P’s of Research**—Purpose, Paradigm, Process, Participants, Products, and Presentation—form a comprehensive framework that guides every stage of a research project. This model ensures that a study is not only systematic but also ethical, rigorous, and impactful.

1. Purpose

The **Purpose** identifies the core "why" of the project, including the specific topic, research questions, and set objectives. Research without a clearly defined purpose is unlikely to be of high quality.

- **Role in Shaping Project:** It acts as a roadmap, justifying why a topic is worthwhile and ensuring the study contributes new knowledge rather than merely repeating existing work.
- **IT Example:** A researcher might undertake a study to discover the most effective **phishing website detection techniques** to assist security managers in selecting the best anti-phishing approaches.

2. Paradigm

A **Paradigm** is a shared way of thinking or a philosophical framework that guides how a researcher perceives reality and knowledge.

- **Role in Shaping Project:** It provides the underlying assumptions for the study. For instance, **Positivism** emphasizes objectivity and measurable facts, whereas **Interpretivism** focuses on understanding subjective human experiences.
- **IT Example:** The **survey strategy** is frequently associated with a positivist paradigm because it looks for broad patterns and generalisations across a population.

3. Process

The **Process** refers to the sequence of activities performed, including establishing a conceptual framework, selecting a research strategy (e.g., experiment, case study), and choosing data analysis methods.

- **Role in Shaping Project:** It provides the "master plan" for the investigation. A systematic process ensures the research is valid and acknowledges its boundaries and limitations.
- **IT Example:** A **Systematic Literature Review (SLR)** follows a rule-based process involving specific steps: constructing research questions, identifying electronic databases (like IEEE Xplore), and applying strict inclusion/exclusion criteria.

4. Participants

Participants include everyone directly involved in the study (such as interviewees) and those indirectly involved (such as the academic community, editors, or end-users).

- **Role in Shaping Project:** Researchers must deal with all participants **legally and ethically**. This involves obtaining **informed consent**, protecting the right to **anonymity**, and ensuring no harm comes to those involved.
- **IT Example:** In a study on user satisfaction with an automation system, participants might be selected via **purposive sampling**. Ethical considerations require that their personal data be stored securely using encryption.

5. Products

Products are the outcomes of the research, representing the tangible and intangible contributions to the "knowledge frontier".

- **Role in Shaping Project:** Products must answer the original research questions and provide evidence-based conclusions. They often lead to further investigation by identifying unexplained horizons.
- **IT Example:** Tangible products include a final **thesis**, a peer-reviewed **journal article**, or a **technical report**; they can also include new datasets, such as the *PhiKitA* dataset for phishing detection.

6. Presentation

Presentation is the final stage, focusing on effectively communicating and disseminating results to the academic community and stakeholders for scrutiny.

- **Role in Shaping Project:** Professionalism is vital; if findings are not conveyed clearly through accepted academic styles, the audience may assume the entire research project was conducted unprofessionally.
- **IT Example:** Presentation formats in computer engineering include **conference talks**, **thesis defenses**, **posters**, and **software demonstrations** to clients or examiners.

5. Is research merely an accidental discovery? Discuss the concept and meaning of research, and explain the six P's of research.

Research is not merely an accidental discovery; rather, it is a **systematic, planned, and managed process** designed to ensure the information generated is credible. While accidental discoveries can occur within a structured research environment—often taking the form of a phenomenon not previously noticed—they usually lead to a more formal, structured process to verify or understand the observation.

The Concept and Meaning of Research

Research is fundamentally defined as the **"systematic approach to obtaining and confirming new and reliable knowledge"**. It is an orderly procedure that applies a series of steps to increase our understanding of a topic or issue. Key definitions from various perspectives include:

- **General Perspective:** A systematic inquiry used to describe, explain, predict, and control observed phenomena.
- **Academic Perspective:** A careful investigation or inquiry, especially through the search for new facts in any branch of knowledge.
- **Scientific Perspective:** A **controlled, empirical, and critical investigation** of natural phenomena guided by theory and hypotheses regarding presumed relations.
- **Intellectual Endeavour:** A process that has developed over centuries, always searching for truth and aiming to discover, develop, and verify knowledge.

Ultimately, research is about **extending the "knowledge frontier"** by building upon existing work to solve real-world problems. It is often described as a **circular process**, as the interpretation of findings frequently leads to new questions, starting the research cycle over again.

The 6 P’s of Research

The **6 P’s of Research** provide a comprehensive framework for understanding the essential components that shape a successful research project:

1. **Purpose:** This refers to the core **reason for undertaking the study**, including the topic of interest, its importance, the objectives set, and the specific research questions asked. Without a clearly defined purpose, research is unlikely to be of high quality.
2. **Paradigm:** A paradigm is a **shared way of thinking** or a philosophical framework that guides how the researcher perceives reality and knowledge. Examples include **Positivism** (focusing on objectivity and measurable facts), **Interpretivism** (focusing on subjective human experience), and **Critical Research** (challenging societal power dynamics).
3. **Process:** This is the **sequence of activities** undertaken, including identifying the topic, establishing a conceptual framework, selecting a research strategy (e.g., experiment, survey, or case study), and choosing data analysis methods. A systematic process ensures the research is valid and acknowledges its boundaries.
4. **Participants:** This includes everyone **directly involved** (such as interviewees and observed subjects) and those **indirectly involved** (such as editors, the academic community, and end-users). Success depends on dealing with all participants **legally and ethically**, ensuring their rights to anonymity, confidentiality, and informed consent are protected.
5. **Products:** These are the **outcomes of the research**, representing the tangible and intangible contributions to knowledge. Products can include a **final thesis, peer-reviewed journal articles, conference papers, or computer-based products** like new software or datasets.
6. **Presentation:** This is the means by which research is **disseminated and explained to others**. It involves communicating findings through accepted academic styles, such as oral defenses, conference talks, or posters. Professional presentation is vital; otherwise, the audience may assume the entire project was conducted unprofessionally.

6. Creativity in the research process

Research is fundamentally a **creative process** that extends beyond simple logic to include insight, intuition, and curiosity. While the structured, logical steps of research can be taught, the "art" of research—characterized by imagination and initiative—is a skill researchers must cultivate through practice.

The Role of Creativity in Research

Creativity is often what distinguishes **outstanding research** from merely satisfactory work. In fields like computer engineering, it manifests through **innovation** (developing new algorithms) and **optimisation** (improving existing systems). Within the research process, creativity is essential for:

- **Generating Ideas:** A good research idea often requires thinking "outside the box" or taking a completely different approach from previous scholars.

- **Problem Solving:** Addressing real-world challenges often requires a creative synthesis of existing knowledge to find new applications.
- **Qualitative Writing:** Qualitative research designs particularly value creativity, allowing for innovative frameworks and a more literary, descriptive writing style.

Fostering Creativity (Ladd, 1987)

Creativity can be nurtured through specific habits and attitudes. Key strategies for fostering a creative research environment include:

- **Writing as Thinking:** Frequently, researchers do not fully understand their own perspectives until they **write them down**. The act of writing helps clarify thoughts and uncover new relationships.
- **Tolerance for Uncertainty:** Researchers must be willing to "grope" and ponder through unexplained horizons of knowledge, cultivating a tolerance for situations where the path is not immediately clear.
- **Challenging Assumptions:** Outstanding research often involves questioning long-held beliefs or looking at existing problems in **alternate ways**.
- **Generating Abundant Ideas:** Most ideas are "bad" ideas, but having many ideas is better than having none. Researchers should generate a high volume of ideas—a **breadth-first approach**—before filtering them through critical analysis.
- **Incubation Periods:** Creativity is often sparked by setting a problem aside and returning to it later, as well as ensuring **freedom from distraction** to provide dedicated time to think.
- **Collaboration:** Exchanging ideas with colleagues and using previously developed knowledge as a foundation can provide the spark for new insights.

Creativity and the Research Cycle

The research process is **circular**, meaning findings often lead to new, creative questions that start the cycle again. By combining rigorous systematic procedures with creative intuition, researchers can push the "knowledge frontier" forward.